Band-1 Hadamard Order 3908 — Miyamoto (Double-seed H_976) + Paley-I; Gap 2052 Still OPEN

H_488/H_976/C_978/H_3908 PASS; 3908 in SPK via Paley-I; gap 2052 still OPEN; conjecture OPEN; GATE-04 LOCKED

C.M. Baird; ZoraASI mission-control Band-1 dash

2026-09-15

Document class. Scientific computational / construction certificate (Band-1).
Program. Open-mathematics collapse list · Human+AI attack order · ZoraASI mission-control.
Parents. `2026-09-15_Band1_Hadamard_3748_Miyamoto`; `2026-09-15_Band1_Hadamard_3188_Miyamoto` (Miyamoto + Paley-I SPK pattern).
Lab. `/workspace/experiments/band1/hadamard_3908_miyamoto/`.
Cutoff. 2026-09-15T13:15:04Z (~9:15AM ET).

Honesty stamp (non-negotiable).
This memo claims **finite constructions of orders 488, 976, and 3908 only** (verified $\max|HH^\top - nI| = 0$), via Miyamoto with Double-seed H_{976} = Sylvester-double(H_{488}) where H_{488} is Paley type I ($q = 487$), and independently via Paley type I. It **does not** claim the full Hadamard conjecture. It **does not** stamp Clay-closed or finished-ToE. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved / Hadamard-conjecture-solved claims. Standard constructive gap **2052** remains **OPEN** under Sylvester∪Paley∪Kronecker (SPK) closure. Order 3908 **enters SPK via Paley-I** ($q = 3907$ prime $\equiv 3 \pmod 4$; order $q + 1 = 3908$): Paley-II **FAIL** ($q = 1953$ not prime); Kronecker **FAIL** ($3908 = 4 \cdot 977$, 977 not Hadamard-admissible). Seed H_{976} uses Sylvester double of Paley-I ($q = 487$); Paley-II seed for 976 **FAIL** ($q = 487 \equiv 3 \pmod 4$). `clay_closed = false`.

Abstract

Prior Band-1 ships constructed Miyamoto-eligible orders through 3748 on box. This increment ships **order 3908** ($3908 = 4 \cdot 977$, 977 prime $\equiv 1 \pmod 4$) via the **Double-seed** pattern plus a classical **Paley-I SPK path** (mirroring 3188/3748 Double-seed, with Paley-I rather than Paley-II for SPK). Seed H_{488} via Paley type I ($q = 487$); H_{976} = Sylvester-double(H_{488}); symmetric Paley conference C_{978} over prime 977; H_{3908} = Miyamoto(H_{976}, C_{978}). Independently, H_{3908} via Paley type I ($q = 3907$). Unit tests **28/28 PASS**. SPK next gap remains **2052** (OPEN). Order 3908 **is in SPK** via Paley-I (not Miyamoto-only). The **full Hadamard conjecture remains OPEN**.

Keywords. Hadamard matrix; order 3908; Miyamoto; Sylvester doubling; Paley type I; Paley conference; constructive gap 2052; SPK; Band-1; non-claim.

1. Statement lock

Definition. A Hadamard matrix of order n is an $n \times n$ matrix H with entries in ± 1 satisfying $HH^\top = nI$.

Finite lemmas (this increment). 1. H_{488} constructed and verified (Paley type I, $q = 487$). 2. H_{976} constructed and verified (Sylvester double of H_{488}). Paley-II seed for 976 **rejected**. 3. C_{978} constructed and verified (Paley conference, $q = 977$): $CC^\top = 977I$, zero diagonal. 4. H_{3908} constructed and verified (Miyamoto). **Not** the Hadamard conjecture. 5. H_{3908} also constructed and verified (Paley type I, $q = 3907$) — SPK classical path. 6. SPK: $3908 \in$ SPK via Paley-I. Paley-II/Kronecker paths for 3908 **rejected**. Next constructive gap remains **2052** (OPEN).

2. Arithmetic and eligibility

Quantity	Value
Target order	$3908 = 4 \cdot 977$
$q = 977$	prime, $977 \equiv 1 \pmod 4 \rightarrow$ Miyamoto-eligible
Needed H_{q-1}	H_{976}
Needed conference	order $978 = q + 1$
H_{488} path	Paley type I, $q = 487$ (prime $\equiv 3 \pmod 4$)
H_{976} path (used)	Sylvester double of H_{488}
H_{976} Paley-II	**REJECTED**: $2(q' + 1) = 976 \Rightarrow q' = 487$ prime but $\equiv 3 \pmod 4$
Conference path	Symmetric Paley over prime 977
Paley-I for 3908	**PASS**: $q' = 3907$ prime $\equiv 3 \pmod 4$; $q' + 1 = 3908$
Paley-II for 3908	**REJECTED**: $q' = 1953$ not prime
Kronecker for 3908	**REJECTED**: only 4×977; $977 \not\equiv 0 \pmod 4$ (not admissible)

3. Construction results

Object	Method	$\max \lvert HH^\top - nI \rvert$ (or conf)	Wall (s)	Status
H_{488}	Paley type I, $q = 487$	0	0.1722	**PASS**
H_{976}	Sylvester double of H_{488}	0	0.428	**PASS**
C_{978}	Paley conference, $q = 977$	0	0.8697	**PASS**
H_{3908}	Miyamoto(H_{976}, C_{978})	0	25.5009	**PASS**
H_{3908}	Paley type I, $q = 3907$	0	33.146	**PASS**

Primary pipeline wall (incl. unit tests + SPK refresh + Paley-I) $\approx$ 62.244 s. Compressed ± 1 matrices saved as `H488_pm1.npz`, `H976_pm1.npz`, `C978_pm1.npz`, `H3908_pm1.npz`, `H3908_paleyI_pm1.npz` (int8).

4. SPK closure note

Seeds: all admissible $\leq 2000 \cup$ on-box prior classical/SPK ships (through 3748/936/468/3316/3188/…; **exclude** 3716/3524/3284/3076/3044/2836/2692/2564 Miyamoto-only) $\cup$ verified H_{488}, $H_{976} \cup$ Paley-I order 3908 $\cup$ Sylvester $\cup$ Paley I/II, closed under Kronecker (limit 4000). **Miyamoto is not folded into SPK**; order 3908 enters SPK by the classical Paley-I path.

Quantity	Value
Known count	880
3908 in SPK	**True** (via Paley-I)
976 in SPK	**True** (Sylvester double of Paley-I)
488 in SPK	**True** (Paley-I)
Next constructive gap	**2052** (OPEN)
Miyamoto-only bonus?	**No** (in SPK via Paley-I; like 3188)

5. Gap 2052 (OPEN; method triage / non-finding ONLY)

Path	Result
Paley-I ($q = 2051$)	FAIL
Paley-II ($q = 1025$)	FAIL
Kronecker	FAIL ($2052 = 2^2 \cdot 3^3 \cdot 19$)
Miyamoto + Paley conference ($q = 513$)	FAIL (513 not prime)
GS / SDS / Williamson / Yamada	lit-only; not constructed this run

Status: OPEN. Non-finding $\neq$ counterexample. Conjecture remains OPEN.

6. Unit tests

28/28 PASS (Sylvester/Paley/Miyamoto controls; $q = 487/977/3907$ arithmetic; Double path for H_{976}; Paley-II seed path for 976 rejected; SPK path eligibility for 3908 via Paley-I; reject-all-ones; conference control; gap-2052-not-filled guard).

7. Non-claims (explicit)

- No full Hadamard conjecture claim.
- No Clay-closed stamp; `clay_closed = false`.
- No finished-ToE claim; GATE-04 **LOCKED**; no invented J.
- No RH / P≠NP / YM / Frankl / NS-solved / AC-solved claims.
- Gap **2052** remains **OPEN** under SPK.
- Order 3908 is **in SPK via Paley-I** (Miyamoto also verified; not Miyamoto-only).

8. Next optional

method-attack SPK gap **2052** OR next Miyamoto ($q = 1009 \to n = 4036$ Double-seedable; or $q = 997 \to n = 3988$) OR Track B privacy / Object 1 meter await.

Lab: `/workspace/experiments/band1/hadamard_3908_miyamoto/` · *Script:* `verify_hadamard_3908.py` · *Artifacts:* `results.json`, `SUMMARY.txt`, `literature_ledger.json`, `*.npz`.